BIFlow BO3 Data Quality Report

Full cleaning run summary generated from on-disk artifacts.

Dataset	titanic
Run id	upload_mydata_20261026_103233
Generated	2026-10-26 09:00 UTC
Source	titanic.csv (4a437fde05fe...)
Shape	891 × 12
LLM	LLM off
Config hash	e7bebfb8b1e85efd
Versions	Python 3.14.6, pandas 3.1.0, sklearn 1.9.1, reportlab 5.1.0

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Executive summary

PASS

Rows in	891	Rows out	891
Duplicates removed	0	Cells imputed	179
Outliers flagged	182	Review items	0
Silent deletions	0	Runtime ms	—

0 exact duplicate rows were removed and logged where applicable.
179 cells were imputed using configured deterministic or ML methods.
182 outlier flags were applied on numeric columns per YAML thresholds.
0 items remain in the review queue for human decision.

Safety & integrity checks

Invariant checks derived from parquet files and logs (never hardcoded).

Check	Status	Evidence
Row accounting titanic	PASS	in=891 dropped=0 out=891
Dtype preservation titanic	PASS	all original dtypes compatible
No literal 'nan' strings titanic	PASS	none
Undo round-trip	NOT RUN	Full undo verification requires playground helper
Determinism	NOT RUN	No run_snapshot.json
Ingest sha256 recorded	PASS	titanic.csv 4a437fde05fe…

Input & parsing

Global parse settings: {"encoding": "", "delimiter": ",", "header": true, "decimal": ".", "thousands": null, "dayfirst": true, "na_tokens": ["", "NULL", "N/A", "null", "None", "nan", "?", "-", "NaN", "#N/A", "NA"]}

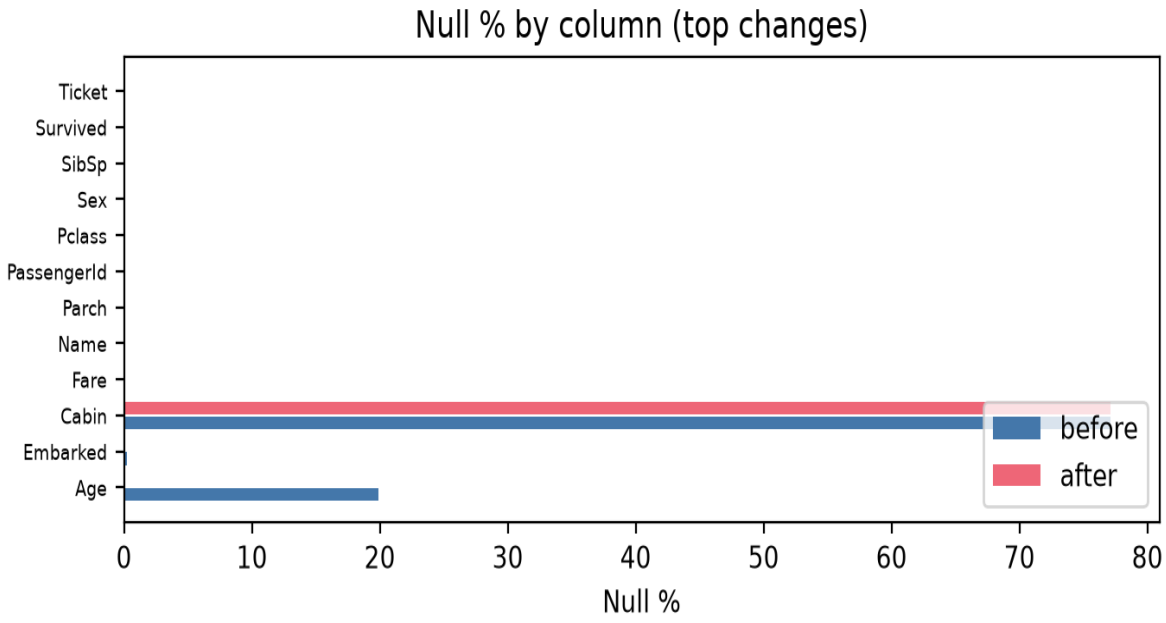
File	Encoding	Delimiter	Rows	Cols	Warnings
titanic.csv	auto	,	891	12	Column 'PassengerId': id-like (near-unique integers (891 uni

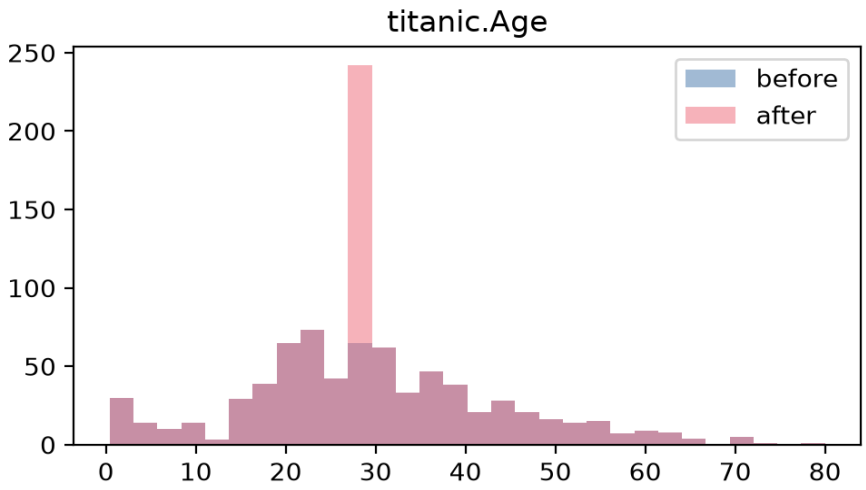
Column recognition

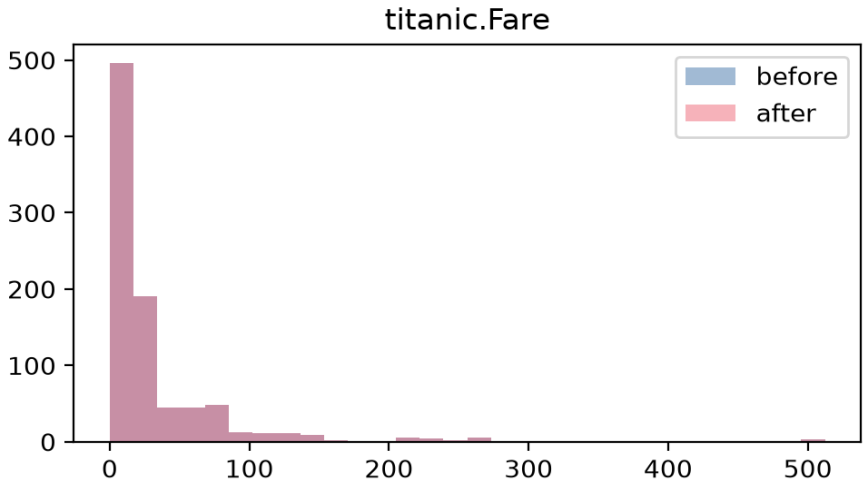
Table	Column	Role	Type	Conf.	Action	Evidence
titanic	PassengerId	id	identifier_integer	0.92	none	id role + numeric
titanic	Survived	binary_flag	boolean_flag	0.95	standardise_safe	two-value flag column
titanic	Pclass	categorical	categorical	0.85	review	low/medium cardinality (3 values)
titanic	Name	id	identifier_alphanumeric	0.93	none	id role + letters in values
titanic	Sex	categorical	categorical	0.85	review	low/medium cardinality (2 values)
titanic	Age	continuous_numeric	continuous_numeric	0.88	review	continuous numeric role
titanic	SibSp	categorical	categorical	0.85	review	low/medium cardinality (7 values)
titanic	Parch	categorical	categorical	0.85	review	low/medium cardinality (7 values)
titanic	Ticket	id	identifier_alphanumeric	0.93	none	id role + letters in values
titanic	Fare	continuous_numeric	currency	0.88	review	currency-like strings in money column
titanic	Cabin	categorical	categorical	0.85	review	low/medium cardinality (147 values)
titanic	Embarked	categorical	categorical	0.85	review	low/medium cardinality (3 values)

Profile before vs after

Table	Rows	Dupes	Null cells (sum)
titanic	891	0	866







Cleaning actions

Rule	Method	Columns	Rows affected
detect_outliers	iqr	Age, Fare	182
impute_categorical	ml	Embarked	2
impute_numeric	ml	Age	177
null_strategy	deterministic	Cabin	687

Table	Column	Rule	Rows	Before	After
titanic	Cabin	null_strategy	687	—	—
titanic	Age	impute_numeric	177	—	—
titanic	Embarked	impute_categorical	2	—	—
titanic	Age	detect_outliers	66	—	—
titanic	Fare	detect_outliers	116	—	—

Review queue

Nothing needs review

Dropped rows & reconciliation

Table	Rows in	Dropped	Rows out	Status
titanic	891	0	891	PASS

Lineage

Step	Method	Rule	Rows
null_strategy	deterministic	null_strategy	687
impute_numeric	ml	impute_numeric	177
impute_categorical	ml	impute_categorical	2
detect_outliers	iqr	detect_outliers	66
detect_outliers	iqr	detect_outliers	116



LLM usage

All counts and statistics in this report are read from run artifacts; they never depend on LLM output.

LLM not used

Reproducibility & limitations

Known limitations

- Ambiguous dates go to the review queue; no silent guess.
- No semantic correction of free-text values beyond configured rules.
- Column roles block unsafe numeric imputation on ids and categoricals.
- LLM advisory (when enabled) does not change deterministic counts in this report.
- See README for supported upload formats and issue classes.

Right-to-left scripts (e.g. Arabic) are not shaped; glyphs appear in logical order only.

config_used.yaml (first 120 lines):

```
name: uploaded_mydata
domain: upload
source:
type: csv
path: data/uploads/uploaded_mydata
tables:
titanic: titanic.csv
expected_schema:
titanic:
PassengerId:
dtype: int
role: id
why: near-unique integers (891 unique)
Survived:
dtype: int
role: binary_flag
why: binary 0/1
Pclass:
dtype: int
role: categorical
why: categorical, 3 values
Name:
dtype: str
role: id
why: alphanumeric labels with letters
Sex:
dtype: str
role: categorical
why: categorical, 2 values
Age:
dtype: float
role: continuous_numeric
why: continuous numeric (88 distinct)
SibSp:
dtype: int
role: categorical
why: categorical, 7 values
Parch:
dtype: int
role: categorical
why: categorical, 7 values
Ticket:
dtype: str
role: id
why: alphanumeric labels with letters
Fare:
dtype: float
role: continuous_numeric
why: continuous numeric (248 distinct)
Cabin:
dtype: str
role: categorical
why: mixed text, 147 values
Embarked:
dtype: str
role: categorical
why: categorical, 3 values
relationships: []
cleaning_rules:
primary_keys:
titanic: PassengerId
role_inference:
high_null_pct: 60
parse_dates: []
trim_strings:
titanic:
- Name
- Sex
- Ticket
- Cabin
- Embarked
drop_exact_duplicates:
- titanic
null_strategy:
titanic.PassengerId:
```

strategy: flag_only
flag_column: PassengerId_missing_flag
titanic.Survived:
strategy: flag_only
flag_column: Survived_missing_flag
titanic.Pclass:
strategy: mode
imputed_flag: Pclass__imputed
titanic.Name:
strategy: flag_only
flag_column: Name_missing_flag
titanic.Sex:
strategy: mode
imputed_flag: Sex__imputed
titanic.Age:
strategy: median
allowed:
- median
imputed_flag: Age__imputed
titanic.SibSp:
strategy: mode
imputed_flag: SibSp__imputed
titanic.Parch:
strategy: mode
imputed_flag: Parch__imputed
titanic.Ticket:
strategy: flag_only
flag_column: Ticket_missing_flag
titanic.Fare:
strategy: median
allowed:
- median
imputed_flag: Fare__imputed
titanic.Cabin:
strategy: flag_only
flag_column: Cabin__high_null
titanic.Embarked:
strategy: mode
imputed_flag: Embarked__imputed
impossible_values:
- table: titanic
column: Fare
rule: '>= 0'
action: flag
flag_column: Fare__invalid

Appendix

Glossary

Term	Meaning
Role	Inferred column purpose (id, categorical, numeric, ...)
Flag	Boolean helper column (__imputed, __outlier, ...)
Imputation	Fill missing values with a documented statistic or model
Outlier	Values flagged by IQR or similar; not always removed
Review queue	Items needing human approval before auto-apply
Lineage	Ordered record of cleaning steps and row counts